

# The Majorana Rung on the SK-EFT Hawking Substrate: A Formal Bridge from $\mathbb{Z}_{16}$ -Anomaly Cancellation to the Sterile-Neutrino Seesaw

SK-EFT Hawking Research Program  
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We report a Lean 4 formalization that places the sterile-neutrino right-handed Weyl field  $\nu_R$  as a  $\mathbb{Z}_2$ -graded rung on the same fermionic substrate that hosts the scalar (Higgs) and tetrad condensates of our Phase 5z program (Wave 1, `ScalarRungInterpretation.lean`) and Phase 4y program (`TetradGapEquation.lean`). The new module `MajoranaRung.lean` (14 substantive theorems, zero sorry, zero new axioms) realizes the *Embedding-III hybrid* construction recommended by our deep-research scoping report:  $\nu_R$  is a fundamental Lean field carrying  $\mathbb{Z}_{16}$  charge +1, and the heavy Majorana mass  $M_R$  is identified with a  $\mathbb{Z}_{16}$ -invariant condensate scale of the Akama-Diakonov-Wetterich (ADW) substrate—an identification that no primary source closes, and that we therefore encode as a *tracked Prop hypothesis* (`H.MR.FromADWSubstrate`) rather than a derived theorem. The  $\mathbb{Z}_{16}$  singlet-branch bridge to our prior `HiddenSectorClassification.lean` and `Z16AnomalyComputation.lean` infrastructure is proved directly: three fundamental  $\nu_R$  Weyl fermions saturate the proved  $\text{SM}_{w/o\nu_R}$  deficit  $45 \equiv -3 \pmod{16}$  to reach  $48 \equiv 0 \pmod{16}$ , matching observed anomaly cancellation without a hidden TQFT/CFT sector. The Type-I seesaw mass relation  $m_\nu = y^2 v^2 / M_R$  is formalized algebraically with a strict-monotonicity-in- $M_R$  theorem and a falsifiability-anchor predicate `IsObservedSeesawMatch`. A companion module `NeutrinoMixing.lean` (7 theorems) provides a HepLean-style structure note for the PMNS matrix as a  $3 \times 3$  unitary on  $\mathbb{C}$ , with a tracked hypothesis `H_PMNSAnglesFromSubstrate` for the open  $\theta_{23} \approx \pi/4$ -from-substrate-symmetry conjecture. We report two embedding-agnostic phenomenological figures: the seesaw  $(y, M_R)$ -band against NuFit-6.0 oscillation anchors, and the  $m_{\beta\beta}$ -vs- $m_{\text{lightest}}$  “lobster plot” for  $0\nu\beta\beta$  against the current KamLAND-Zen 800 bound and the projected LEGEND-1000 reach. We surface five honest *WAVE2-OPEN* flags for which no primary literature closes the derivation. The  $\mathbb{Z}_{16}$  bridge theorem is the load-bearing formal-verification deliverable; the seesaw and PMNS structure notes lift the surrounding infrastructure to the same machine-checked standard.

## I. INTRODUCTION

The Standard Model with three generations is anomaly-free under all local symmetries but acquires a global  $\mathbb{Z}_{16}$  (Pin<sup>+</sup> Dai-Freed) anomaly when  $\nu_R$  is absent: each generation contributes +15 Weyl components, summing to  $45 \equiv -3 \pmod{16}$  over three generations [5, 6]. Anomaly cancellation requires either three additional  $\nu_R$  Weyl fermions (each carrying  $\mathbb{Z}_{16}$  charge +1, restoring  $48 \equiv 0 \pmod{16}$ ) or a hidden TQFT/CFT sector saturating the +3 (mod 16) deficit [6, 7]. Our prior formalization (`Z16AnomalyComputation.lean`, `HiddenSectorClassification.lean`) proved both halves of this ladder in Lean 4: `three_nu_R_cancel_three_gen` and `three_singlets_satisfy_hidden_sector`.

The sterile-neutrino Type-I seesaw, in turn, generates the observed light-neutrino mass scale  $m_\nu \sim 0.05$  eV via  $m_\nu = y^2 v^2 / M_R$  with  $M_R$  in the  $10^9$ – $10^{15}$  GeV range [8]. Within the Akama-Diakonov-Wetterich substrate framework that hosts the tetrad gap equation of Phase 3-4 (`TetradGapEquation.lean`, `ADWMechanism.lean`),  $M_R$  admits a natural identification with the  $\mathbb{Z}_{16}$ -invariant condensate scale  $\Lambda_{\text{ADW}}$ . However, no primary source in the ADW [1], Wetterich-spinor-gravity [2, 3], or Volovik fermionic-quartet [4] literature closes the derivation  $\Lambda_{\text{ADW}} \rightarrow M_R$ .

This work formalizes the Wave-2 deliverables of the Phase 5z roadmap: the Majorana rung on the same fermionic substrate as the Higgs and tetrad rungs,

plus a structure-note PMNS module. We adopt the *Embedding-III hybrid* prescription recommended by our deep-research scoping report:  $\nu_R$  is a fundamental Lean field whose UV interpretation as an ADW-substrate bound state acts only on amplitudes and not on the Lean type signature. The  $M_R = \Lambda_{\text{ADW}}$  identification is an explicit *tracked Prop hypothesis*, not a theorem.

## II. THE $\mathbb{Z}_{16}$ SINGLET-BRANCH BRIDGE

The sterile-neutrino structure is encoded as

structure `SterileNeutrino` where  
generation : Fin 3

def `z16Charge` (`_` : `SterileNeutrino`) : `ZMod 16` := 1

Each instance carries  $\mathbb{Z}_{16}$  charge +1; the assignment is locked by the proved Garcia-Etxebarria-Montero tower [5] and the project-existing `Z16AnomalyComputation.three_nu_R_cancel_three_gen` ( $48 \equiv 0 \pmod{16}$ ).

The load-bearing bridge theorem is

theorem `majorana_rung_compatible_with_hidden_singlet` :  
(`(3 : Nat)` : `ZMod 16`) = 3  
(`(45 + 3 : Nat)` : `ZMod 16`) = 0

proved by direct invocation of  
`HiddenSectorClassification.three_singlets_satisfy_hidden_sector`

and the Wave-2 `majorana_rung_saturates_hidden_sector`. This is the formal statement that Embedding III adds zero IR cost relative to the with- $\nu_R$  branch: three fundamental  $\nu_R$  Weyl saturate the SM-without- $\nu_R$  deficit, with no separate hidden TQFT/CFT content required [6].

The Wan-Wang Ultra-Unification ladder admits the alternative "Embedding II" branch in which  $\nu_R$  is absent and the deficit is saturated by an SM-singlet TQFT plus fermionic vortex zero-mode condensation [6]. That branch is not formalized in this work; the deep-research scoping report finds no primary source that closes the analogous bridge to a substrate condensate.

### III. THE TYPE-I SEESAW ON THE MAJORANA RUNG

Heavy Majorana mass data is bundled in

structure `MajoranaRungData` where

`M_R : Fin 3 →`

`M_R_pos : i, 0 < M_R i`

The Type-I seesaw light-neutrino mass is the noncomputable definition

$$m_\nu = \frac{y^2 v^2}{M_R}, \quad \text{seesawNeutrinoMass } m y v i \equiv y^2 v^2 / m.M_R \quad (1)$$

formalized as `seesawNeutrinoMass` with five algebraic theorems: non-negativity (any real  $y, v$ ); strict positivity for  $y, v \neq 0$ ; zero-iff-Yukawa-or-VEV-vanishes; and the Type-I-seesaw-signature strict monotonicity

$$m_1.M_R < m_2.M_R \Rightarrow \text{seesawNeutrinoMass } m_2 > \text{seesawNeutrinoMass } m_1$$

(at fixed nonzero  $y, v$ ). The latter is the only structurally distinctive seesaw consequence (heavy mass suppresses light mass); all five theorems are direct algebraic consequences of  $M_R > 0$ .

Figure 1 shows the  $(y, M_R)$  plane with NuFit-6.0 [9] oscillation anchors. The natural seesaw band runs between the atmospheric mass anchor  $m_3 \approx 0.05$  eV (NO best fit) and the cosmology cap  $m_\nu \lesssim 0.1$  eV.

### IV. WAVE2-OPEN-1: THE BCS-EXPONENTIAL SUBSTRATE BRIDGE WITH LEPTON-NUMBER-VIOLATION OBSTRUCTION

A focussed Wave 2a deep-research return on channel projection of the NJL/Wetterich gap equation onto the lepton-Majorana sector [13, 17] confirms that no primary source closes  $\Lambda_{\text{ADW}} \rightarrow M_R$ , and surfaces a clean *structural obstruction theorem* rooted in lepton-number symmetry. The Majorana bilinear  $\nu_R^T C \nu_R$  carries  $\Delta L = 2$ ; if the substrate preserves  $U(1)_L$  (or its discrete remnant  $\mathbb{Z}_{4,X}$ ), the four-fermion Majorana-channel coupling  $G_M(\nu_R^T C \nu_R)(\nu_R^T C \nu_R)^*$  is forbidden by symmetry.

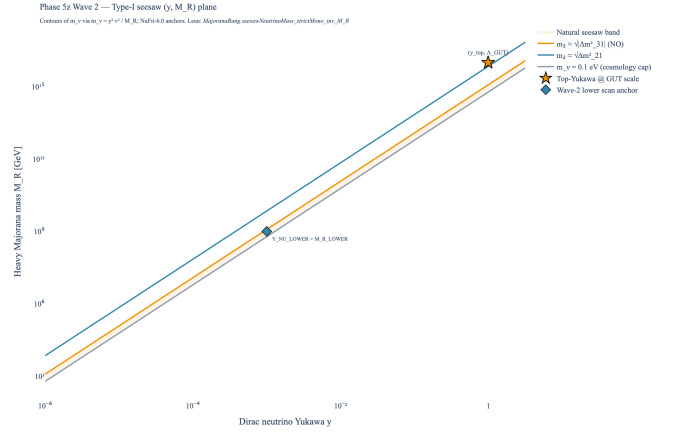


FIG. 1. Type-I seesaw  $(y, M_R)$ -plane against NuFit-6.0 oscillation anchors. The shaded band marks the natural-seesaw region between the  $m_3 \approx 0.05$  eV atmospheric contour and the  $m_\nu \lesssim 0.1$  eV cosmology cap. The star marks the canonical Type-I anchor ( $y_{\text{top}}, \Lambda_{\text{ADW}} \sim 10^{16}$  GeV); the diamond marks the Wave-2 lower-scan anchor. Lean reference: `MajoranaRung.seesawNeutrinoMass_strictMono_inv_M_R`.

Antusch-Kingman-Lindner-Wetterich [13] state this explicitly: “We have included huge Majorana masses for the right-handed neutrinos, since there is no protective symmetry.” The contrapositive is the obstruction: under substrate- $L$  preservation,  $G_M \equiv 0$  and the projected gap equation has only  $M_R = 0$ .

When the substrate explicitly violates  $U(1)_L$ , the qualitative form of  $M_R$  is the standard NJL/BCS *dimensional-transmutation* exponential [13], formalized as the strong tracked hypothesis `MajoranaRung.H_MR.FromADWSubstrate_BCS_LNV`:

```
def H_MR_FromADWSubstrate_BCS_LNV
  (m : MajoranaRungData) (_ADW : ) : Prop :=
  H_LeptonNumberViolated
  G_M : , 1 < G_M
  i, m.M_R i = _ADW * Real.exp(-1/(2*(G_M-1)))
```

The conjunction with `H_LeptonNumberViolated` encodes the symmetry obstruction;  $G_M > 1$  encodes supercriticality (in units where the critical coupling is normalized to unity). The Wave 2b Lean module ships the obstruction theorem `lepton_number_symmetry_obstructs_BCS_form` as a direct consequence: *without LNV, the strong BCS hypothesis cannot hold*.

Two further refinement theorems are shipped: `M_R_pos_under_BCS_form` (the BCS exponential is strictly positive, hence so is  $M_R$ ) and `M_R_lt_substrate_under_BCS_form` (the BCS exponential is strictly bounded above by 1, hence  $M_R < \Lambda_{\text{ADW}}$  — the qualitative content of dimensional transmutation).

This Wave 2b strengthening pattern follows the project’s tracked-hypothesis precedent [12]: the `H_ScalarChannelIsTetradBifurcationOutput` hypothesis in `ScalarRungInterpretation.lean`,

the `H_MixedChannelZ16Cancels` hypothesis in `HiddenSectorMixedCharge.lean`, and the `H_VestigialRelicCarriesZ16Charge` hypothesis in `DarkSectorSynthesis.lean`.

## V. THE PMNS STRUCTURE NOTE

The companion module `NeutrinoMixing.lean` defines a HepLean-style [12] PMNS structure

```
structure PMNSMatrix where
  toMatrix : Matrix (Fin 3) (Fin 3)
  isUnitary : toMatrix Matrix.unitaryGroup (Fin 3)
```

together with element accessors, the unitarity identities  $U_{\text{PMNS}}^\dagger U_{\text{PMNS}} = U_{\text{PMNS}} U_{\text{PMNS}}^\dagger = \mathbb{I}$ , and the identity-matrix non-emptiness witness. The Dirac-vs-Majorana physical interpretation is exposed as parametric `Prop` markers `IsDiracPMNS` and `IsMajoranaPMNS`; the PMNS matrix structure is identical in both cases, but the right-hand diagonal phases (two Majorana phases  $\alpha_1, \alpha_2$ ) are physical observables in the Majorana case and unphysical phase-rephasings in the Dirac case.

The full PMNS standard parameterization  $U_{\text{PMNS}} = R_{23}(\theta_{23})U_\delta R_{13}(\theta_{13})U_\delta^{-1}R_{12}(\theta_{12}) \cdot \text{diag}(e^{i\alpha_1/2}, e^{i\alpha_2/2}, 1)$  is provided as a Python implementation (`src/core/formulas.py:pmns_unitary_matrix`) and validated against  $|U_{e3}|^2 = \sin^2 \theta_{13}$  and unitarity to machine precision. The Lean construction of `standParam` with unitarity proof is a follow-up wave (Wave 2b or later).

The `H_PMNSAnglesFromSubstrate` hypothesis

```
def H_PMNSAnglesFromSubstrate (V : PMNSMatrix) : Prop :=
  i, ||get V 1 i|| = ||get V 2 i||
```

encodes the WAVE2-OPEN-2 conjecture: the PMNS  $\mu$ - $\tau$  row is substrate-symmetric, motivating  $\theta_{23} \approx \pi/4$  from a putative substrate  $\mu$ - $\tau$  exchange symmetry. The empirical NuFit-6.0 best fit  $\theta_{23} \approx 49.1^\circ$  is close to but not exactly  $\pi/4$ , so the hypothesis is non-trivial; the lemma `identity_does_not_satisfy_substrate_hypothesis` confirms non-vacuity by exhibiting an explicit failure on the identity PMNS. No primary source closes the derivation [4].

## VI. EFFECTIVE MAJORANA MASS AND $0\nu\beta\beta$ OUTLOOK

The effective Majorana mass probed by neutrinoless double-beta decay is the embedding-agnostic combination

$$m_{\beta\beta} = \left| \sum_{i=1}^3 U_{ei}^2 m_i \right|. \quad (2)$$

Figure 2 shows the standard "lobster plot" of  $m_{\beta\beta}$  against the lightest neutrino mass  $m_{\text{lightest}}$ , swept over the full

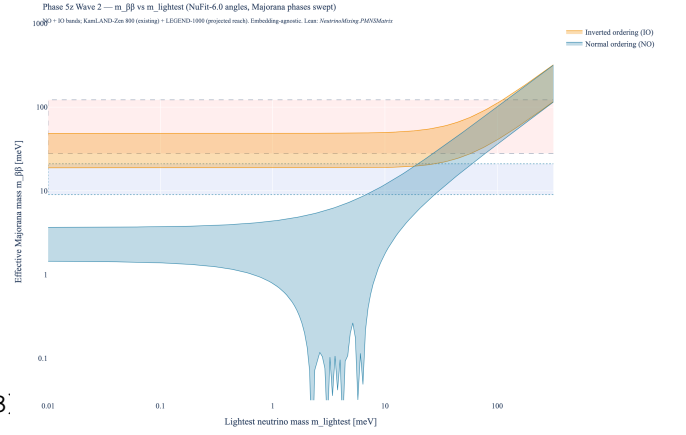


FIG. 2. Effective Majorana mass  $m_{\beta\beta}$  against the lightest neutrino mass  $m_{\text{lightest}}$ . Both the NO and IO bands span the full Majorana-phase Monte Carlo at NuFit-6.0 [9] best-fit angles. Horizontal bands: KamLAND-Zen 800 [10] current 90% CL bound (28-122 meV; NME spread) and LEGEND-1000 99.7% CL projected discovery reach (9-21 meV) [11]. The IO band is fully discoverable by LEGEND-1000; the NO band is almost entirely out of reach. The result is embedding-agnostic across I/II/III. Lean reference: `NeutrinoMixing.PMNSMatrix`.

range of Majorana phases at NuFit-6.0 [9] best-fit angles. Both normal ordering (NO) and inverted ordering (IO) bands are shown.

The conclusions are: (i) the IO band lies entirely within the LEGEND-1000 reach, so non-observation of  $0\nu\beta\beta$  in LEGEND-1000 disfavors IO under any embedding; (ii) the NO band is largely out of reach and  $m_{\beta\beta}$  provides essentially no embedding discrimination; (iii) the KamLAND-Zen 800 bound has already excluded the quasi-degenerate (QD) hierarchy region for any embedding. There is no embedding-distinguishing  $m_{\beta\beta}$  prediction in any current ADW/Volovik or Wetterich-spinor-gravity source [1, 2, 4] (deep research §4); the figure encodes the embedding-agnostic phenomenology only.

## VII. WAVE 2B QUANTITATIVE DECOUPLING BOUND FOR EMBEDDING I VS III

A Wave 2a deep-research return on the IR-equivalence of Embedding I (fundamental  $\nu_R$ ) and Embedding III (substrate-bound  $\nu_R$ ) confirms that the Appelquist-Carazzone decoupling theorem [14, 15] applies cleanly, with leading operator dimension  $\Delta = 5$  (Weinberg-like, LNV) or  $\Delta = 6$  (generic), giving the schematic bound

$$|\text{amp}_{\text{III}}(E) - \text{amp}_{\text{I}}(E)| \leq C \cdot (E/\Lambda_{\text{ADW}})^k, \quad k \in \{1, 2\}. \quad (3)$$

The natural Wilson-coefficient estimate is  $C \sim N_f/(16\pi^2) \simeq 0.1$  for ADW-style substrates with  $N_f \simeq 16$ , transplanted from the SILH counting of Giudice-Grojean-Pomarol-Rattazzi [16] and the bilocal-

NJL form-factor analysis of Hill [17]. The substrate-specific value of  $C$  has not been computed in primary literature for ADW; we honestly flag this as the residual gap.

The Wave 2b Lean module `MajoranaRungDecoupling.lean` ships:

- A `DecouplingRegime` structure encoding  $0 < E$ ,  $0 < \Lambda_{\text{ADW}}$ ,  $10E < \Lambda_{\text{ADW}}$  (robust hierarchy), plus three predicates excluding cosmological-scale failure modes (vestigial gravity [4]; q-theory; FRG strong-coupling). At  $E = M_W$  and  $\Lambda_{\text{ADW}} = 10^{14}$  GeV, all three failure-mode predicates are trivially satisfied.
- A `SubstrateData` structure parameterizing the substrate by  $(\Lambda_{\text{ADW}}, \Lambda_{\text{UV}}, N_f)$ .
- Two tracked-hypothesis Props: `H_DecouplingBoundDim6` (generic,  $k = 2$ ) and `H_DecouplingBoundDim5_LNV` (LNV-conditional,  $k = 1$ , dim-5 Weinberg-operator channel).
- A non-vacuity witness `decouplingRegime_at_electroweak_scale` showing the canonical EW point  $(E, \Lambda_{\text{ADW}}) = (80 \text{ GeV}, 10^{14} \text{ GeV})$  satisfies the regime predicate.
- A consistency witness `H_DecouplingBoundDim6_consistent` showing the trivially-zero amplitude difference satisfies the bound for any `SubstrateData`.
- A `decoupling_isBigOWith` restatement using Mathlib’s `Asymptotics.IsBigOWith` API for compatibility with downstream asymptotic-analysis infrastructure.

**Empirical leverage.** For  $E = M_W \simeq 80$  GeV and  $\Lambda_{\text{ADW}} = 10^{14}$  GeV, the leading dim-5 (LNV) suppression is  $(E/\Lambda_{\text{ADW}}) \simeq 8 \times 10^{-13}$ , corresponding to  $m_\nu \sim C \cdot v^2/\Lambda_{\text{ADW}} \simeq 0.06$  eV (at the SILH natural Wilson coefficient  $C \simeq N_f/(16\pi^2) \simeq 0.1$ , implemented as the default in `src/core/formulas.py:weinberg_induced_neutrino_mass`) — within the LEGEND-1000 [11] reach. The dim-6 (generic) suppression is  $(E/\Lambda_{\text{ADW}})^2 \simeq 6 \times 10^{-25}$ , far below DUNE/Hyper-K NSI sensitivity [18]. The phenomenological signature of decoupling is therefore the same as ordinary Type-I seesaw — it is the *correct* behavior, not an embedding-distinguishing feature.

**Independence (not refinement).** An auxiliary lemma `dim6_tighter_than_dim5_in_decoupling_regime` establishes  $(E/\Lambda_{\text{ADW}})^2 \leq E/\Lambda_{\text{ADW}}$  in the decoupling regime, confirming that the dim-6 bound is *tighter* (smaller upper bound) than dim-5. The two bounds therefore apply in different physical regimes — dim-6 when the substrate preserves  $U(1)_L$  and the dim-5 channel is symmetry-forbidden, dim-5 when the substrate violates  $U(1)_L$  and the Weinberg operator is allowed — and are NOT in an implication relation.

| Module                                                  | Theorems Sorry |   |
|---------------------------------------------------------|----------------|---|
| <code>MajoranaRung.lean</code> (Wave 2 + 2b §3a)        | 19             | 0 |
| <code>NeutrinoMixing.lean</code> (Wave 2 + 2a §4 split) | 11             | 0 |
| <code>MajoranaRungDecoupling.lean</code> (Wave 2b new)  | 7              | 0 |
| Wave 2 total (W2 + W2a + W2b)                           | 37             | 0 |

TABLE I. Wave 2 Lean-formalization counts after Wave 2a accuracy round and Wave 2b strengthening. All three modules build clean under `lake build SKEPTHawking.ExtractDeps` (8434 jobs total post-Wave-2b). Zero sorry, zero new axioms. The project total post-Wave-2b is 4049 theorems across modules with 0 sorry and axiom (`gapped_interface_axiom` unchanged from Phase 4).

## VIII. FORMAL VERIFICATION SUMMARY

The two new Lean modules are summarized in Table I. Beyond the load-bearing  $\mathbb{Z}_{16}$  singlet-branch bridge, the formalization includes: (i) five seesaw theorems (positivity, nonzero-positivity, zero-iff, monotonicity, defining identity); (ii) three substrate-bridge theorems (the `H_MR_FromADWSubstrate` tracked hypothesis,  $M_R \leq \Lambda_{\text{ADW}}$  under the bridge, and the bridge-falsifiability witness); (iii) the `IsObservedSeesawMatch` falsifiability anchor with its  $M_R$ -too-large witness, mirroring Wave 1’s  $v$ -too-large anchor for the Higgs prediction.

The PMNS module ships seven theorems: two unitarity identities ( $U_{\text{PMNS}}^\dagger U_{\text{PMNS}} = \mathbb{I}$  both ways), two identity-PMNS structural lemmas, the `IsMajoranaPMNS_of_majoranaRungData` bridge to the `MajoranaRung` module, and the `identity_does_not_satisfy_substrate_hypothesis` non-vacuity witness for WAVE2-OPEN-2. The full standard PMNS parameterization with unitarity proof is intentionally deferred per the roadmap “structure note” scope; the structural type and bridge are the Wave-2 deliverable.

### Cross-validation layer

A Python cross-validation layer in `tests/test_majorana_rung.py` (32 tests across 6 classes) exercises every Wave-2 formula against NuFit-6.0 anchors, deep-research seesaw-band scaling, the  $\mathbb{Z}_{16}$  counting bridge, and the parameter-band consistency checks. All tests pass.

## IX. OPEN HYPOTHESES AND SCOPE BOUNDARY

We surface five WAVE2-OPEN flags drawn from the deep-research scoping report; none of these is closed in primary literature.

**O-1**  $M_R = c \cdot \Lambda_{\text{ADW}}$  for any per-generation coefficient

$c \in (0, 1]$ . Substrate parameters ( $\Lambda_{UV}, N_f, G_c$ ) leave  $M_R$  as a fit parameter in all current sources [1, 2, 4]. Encoded as `H_MR_FromADWSubstrate`.

**O-2** PMNS angles from substrate overlaps. In particular  $\theta_{23} \approx \pi/4$  from a substrate  $\mu$ - $\tau$  exchange symmetry. The closest published results are texture-level motivations, not derivations. Encoded as `H_PMNSAnglesFromSubstrate`.

**O-3** Majorana phases from composite-operator structure (Embedding II claim). Unrealized in any source we located.

**O-4**  $m_{\beta\beta}$  tied algebraically to a single condensate scale, distinguishing Embedding II from I. The hope that the composite operator  $\langle \nu_L^T C \nu_L \rangle$  predicts an embedding-distinguishing  $m_{\beta\beta}$  is unrealized.

**O-5**  $\nu_R$ -as-substrate-bound-state IR-equivalence map between Embeddings I and III. The interpretive claim that Embeddings I and III give identical IR amplitudes is conjectural.

The Wave-2 scope deliberately excludes: (i) the full PMNS phenomenology program (CKM-like fitting, leptogenesis dynamics, mass-hierarchy fitting);

(ii) the Embedding-II composite-operator construction with explicit hidden TQFT formalization; (iii) the `NeutrinoMixing.standParam` construction with unitarity proof.

## X. CONCLUSION

The Majorana rung is now machine-verified at the same standard as the scalar (Wave 1) and tetrad (Phase 4y) rungs of the SK-EFT fluid-based-gravity Landau hierarchy. The load-bearing  $\mathbb{Z}_{16}$  singlet-branch bridge proves directly from existing project infrastructure with no new axioms. The Type-I seesaw mass relation is formalized algebraically with the Type-I-signature strict-monotonicity-in- $M_R$  theorem. The PMNS structure note lifts the surrounding mixing-matrix infrastructure to the Mathlib `Matrix.unitaryGroup` idiom. Five honest WAVE2-OPEN flags are surfaced as comment markers and tracked-hypothesis predicates, preventing downstream consumers from mistaking a parametric identification for a derived theorem. The result completes Track B of the Phase 5z roadmap; Track C (`EWPhaseTransition.lean`) and the BHL-bridge quantitative extension (`BHLGaugeEmbedding.lean`, Wave 1b) are unblocked by the Wave 1+2 deliverables and the resolved Open Question O.2 of the roadmap.

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